Grothendieck Ring Classification
Current Research Status and Audit Corrections

Fusion Frontier research project

25 October 2026

AI-generated research; independent human verification and formalization pending. All theorem and classification labels describe the project's current claims. Exact finite replays do not prove their external categorical premises, and a verified tensor list alone does not prove enumeration completeness.

1 Current classification scope

Ranks two, three and four have unrestricted classification claims, with two, five and fifteen realized based rings respectively. Higher ranks have the explicit bounds below. Realization means existence of at least one model, not classification of all fusion categories up to tensor equivalence.

Rank	Through multiplicity	Excluded	Realized	Open
5	20	40982	17	0
6	9	25362	57	2
7	5	9976	30	0
8	3	4403	93	4
9	2	2145	76	4
10	1	136	58	0
11	1	155	27	0

There are **10 current generally unresolved cases**. The exact current list and tensors are supplied in `data/current_remaining_cases.py`. Rank six is complete through multiplicity three; rank eight through two; rank nine through one. These finite statements are relative to the supplied census and the recorded categorification decisions.

2 Current theoretical results

The ten open cases are listed in rank order:

- Rank six: `R6_4_155`, `R6_8_1660`.
- Rank eight: `R8_3_928`, `R8_3_941`, `R8_3_1154`, `R8_3_1768`.
- Rank nine: `R9_2_137`, `R9_2_1607`, `R9_2_1975`, `R9_2_1976`.

No new exclusion of one of these source rings is claimed.

Rank ten is complete at multiplicity one. The classical conformal embedding $SU(4)_6 \subset SU(10)_1$ supplies an actual connected etale algebra $B = 000 + 060 + 202 + 222$. The charge-zero component of its module category realizes $R10_1_189$. The proof uses minimal ground-state branching and dimension exhaustion; all 1000 census coefficients agree. An actual rank-fourteen modular category and its algebra are also constructed. This recognizes a known construction; no new tensor category is claimed.

Rank eight retains four open cases. R928 has categorical Morita rigidity without a module-rank cutoff. For R1768 an actual complex twisted $GH(C_6, 2)$ precursor has now been constructed with precisely the required pointed associator. The connected separable algebra $1 + \sigma$ needed for the source ring remains unproved. The canonical inherited algebra is instead $\mathrm{End}(\sigma)$. Its remaining multiplication equations are explicit in two scalar variables. Five pointed-extension construction families for R1154 and further exceptional quantum-group routes are ruled out; those route obstructions do not imply general nonexistence.

Rank nine retains four open cases. The centers have no nontrivial Tannakian subcategory, and their integral simples are invertible. A new module-functor sign theorem gives local Clifford restrictions on each large object in the three commutative sources. An all-cochain D8 theorem gives an aggregate condition for all four. Exact discrete certificates force projected cubic indicators $1/2$ in the positive-TY R137 and negative-TY R1976 branches, hence 9 divides their FS exponents. Exact feasible witnesses to the necessary systems remain; they are not categories.

Rank six retains two open cases. The even cyclic Clifford theorem passes all 1144 tested exact blocks for R155. The R1660–R941 identification is exact, but its categorical Morita link still requires the indicated nonpointed algebra.

The current proofs, independent agent audits and portable replays are in `evidence/unresolved_retry_20261024/`. The thirteen new portable checks pass, also with assertions disabled. They do not replace human verification of the categorical arguments.

3 R436: an independent replacement proof

The old TPE scalar-table completeness premise remains withdrawn. Its complete-table derivation assumes unitarity and does not parameterize an arbitrary original spherical category. Exact scalar computations cannot repair this categorical gap.

A separate spherical-equivariantization and formal-codegree argument now excludes $R8_3_436$ generally, without the unsupported parameterization. It is proved in `evidence/r436_repair_20261024/`. The conservative audit correction and the independent replacement are preserved as distinct research steps; the former is not presented as a valid general proof. Rank eight at exact multiplicity three therefore has 3507 exclusions, 16 unitary models and four open cases, and cumulatively 4403 exclusions, 93 models and four open cases.

4 Other corrections

The rank-five current replay is

```
python evidence/prepublication_20261017/rank5/replay.py --jobs 2
```

It checks all 40999 inputs, 40982 general exclusions and 17 models. The historical standalone

verifier leaves the two inert-prime frontiers unresolved in general and has complete bound seventeen. The discovery example is a bounded attempt that may return unresolved; it is not guaranteed fresh discovery of the stored composite certificate.

The signed rank-five commutativity lemma now explicitly requires the dimension character to be nonzero on every basis element. The pair-product Gram matrix's row count explicitly denotes the remaining central simple objects. The rank-seven bibliography now names P. G. Budinski. The census documentation updates the supported rank range, companion-PDF count and supplied, hash-matched rank-seven producer source; independent full re-enumeration remains separate. Older six-case frontier totals are scoped to their historical lower-rank range.

5 Correctness audit and integrated corrections

The 25 October audit reviewed the eleven current mathematical manuscripts of its pinned input release (657 pages), their operative criteria and model code, and the accepted classification routes. All 1,433,410 supplied rank-three through rank-eleven tensors passed fusion axioms and exhaustive based-isomorphism checks. The current partition is 1,433,022 general exclusions, 378 realized rings and ten unresolved cases. Rank two and the unbounded lower-rank arguments are separate. No current decision reversal was indicated by the audit.

The model-identification arguments for `R6_2_143`, `R8_2_820` and `R8_2_869` now use the exact replacement proofs integrated into the rank-six/eight manuscripts. The first uses the actual canonical module of a Q-system dual. The latter two use known tube blocks, every associative completion, actual third indicators and balancing; no numerical recognition of the unknown tube block is needed. A direct exact completion identifies `R8_3_721` in 14 search nodes without a census-completeness premise. `R8_2_868` uses the exact external full-center theorem, not its generator matrix alone. All cited categorical existence and indicator theorems remain explicit dependencies.

Implementation corrections add the commutativity guard required by the cyclotomic criterion, exact native square-root and frame bounds, and native cache binding to source, compiler and flags. A historical exploratory integer wrapper now derives its terminal status from verified components; its defective branch was not used by a current classification. The R189 auxiliary branching convention and documentation pointers are corrected.

The revised rank-nine replay covers the accepted baseline and phases two through five, with the omitted phase-three certificates supplied separately. The R920 completeness fallback is exact pseudo-Boolean and shares the discovery implementation; an independent eight-bit success is not claimed. Stored group/cohomology narrowing records are verified by the portable command; fresh full regeneration is in the dated audit evidence. The new current-source rank-six replay retains the full original raw stream, retries complete failed workers in the reference backend and runs the exact post-loop checks. Its actual execution receipt is separate from earlier campaigns. In particular, the pre-revision audit's partly log-derived rank-six coverage and its missing raw rows remain explicitly disclosed.

No census search was wholly regenerated as part of this correctness audit, and no proof-assistant certification or independent reproof of every external associator is claimed. Detailed reports, original failures and source-bound receipts are in `evidence/correctness_20261025/`. Independent human verification and formalization remain pending.

6 Reading and reproducing the atlas

The common record directory, unresolved directory and model catalogue offer exact structural filters for integral, weakly integral, simple, perfect, pointed, commutative, self-dual and multiplicity-free rings. The filters concern the underlying ring; they are independent of categorical existence. Rational root isolation decides integrality, while complete generated-subring closure decides simplicity. Aggregate rank–multiplicity counts retain their full recorded populations. Definitions, examples and standard references are on the atlas's theory page.

```
python tools/verify_audit_revision.py
```

These commands reconcile the audit, the independent repair and the accepted unitary model, and validate current publication identities and structural data. They do not claim a new full replay of every historical categorical proof.

https://sebastienpalcoux.github.io/grothendieck-ring-classification/theory.html